Neural Memory and Interval Calibration

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1 Motivation and research question

Financial prediction intervals should adapt to rising volatility without staying wide after a temporary shock. Neural control, score–threshold adaptation, and retrospective updates already address adaptation [1–3]. I will test whether **a neural forecaster's scale response changes the effect of calibration memory at shock entry and exit**. The interaction may depend on architecture and event duration. Longer context does not necessarily imply longer effective memory.

2 Method and evaluation

Probabilistic GRU and causal TCN models trained from scratch will produce scales s_t. With a fixed zero center in controlled simulations, the interval half-width is

$$h_t = s_t \, Q_{1-\alpha}\big(\{|y_i|/s_i\}_{i=t-W}^{t-1}\big), \qquad \alpha = 0.1.$$

Across five matched training seeds, I will cross short/long contexts with short/long score windows, using shared weights within each fit and paired event/no-event trajectories. A frozen grid varies pulse duration, magnitude, tails, and latent stochastic volatility; true scales are evaluation oracles only. Separately varying current and archived scales will distinguish their effects. Controls include EWMA/fitted GARCH and rolling, adaptive, and strongly adaptive calibration [4].

Public ECB reference rates for four EUR-cross currencies provide an application check: training through 2018, validation in 2019–2020, and exploratory evaluation from 2021. Metrics include interval score, width, simulated conditional coverage, and entry/recovery contrasts. Training seeds and paired paths quantify simulation variation; date-block comparisons retain cross-asset dependence. Predictions remain immutable; tuning excludes test data and event labels.

3 Expected contribution and feasibility

The intended contribution is a falsifiable account of *when forecaster response matters for calibration*. Matched models run locally; architecture-dependent and null outcomes remain possible. Validation lag sensitivity and a finite-memory erasure check help distinguish learned response from implementation errors. Reproducible failures may motivate an adaptation module tested on a new holdout; null results will narrow the claim. November covers robustness and ablations, followed by the report and presentation before December 11.

References

[1] R. Li and A. Rodríguez. Neural Conformal Control for Time Series Forecasting. *AAAI*, 2025.

[2] R. Tumu, A. Singh, and R. Mangharam. AdaptNC: Adaptive Nonconformity Scores for Conformal Prediction under Distribution Shift. *arXiv:2602.01629v2*, 2026.

[3] J. Jun and I. Ohn. Online conformal inference with retrospective adjustment for faster adaptation to distribution shift. *Pattern Recognition*, 180:114406, 2026.

[4] A. Bhatnagar, H. Wang, C. Xiong, and Y. Bai. Improved Online Conformal Prediction via Strongly Adaptive Online Learning. *ICML*, 2023.

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